

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

IT-713 (GS)**B.E. VII Semester Examination, June 2020****Grading System (GS)****Automata and Compiler Design****Time : Three Hours****Maximum Marks : 70**

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

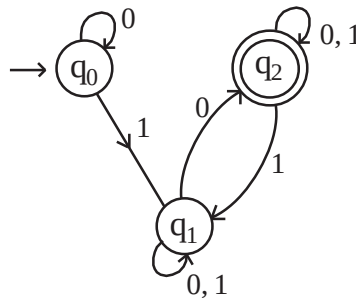
1. Construct DFA for following regular expression use first pos, last pos and follow pos function to construct DFA.

 $a + b^*a + ab$

OR

Construct a transition diagram for recognizing unsigned numbers. Sketch the program segment to implement it.

2. Consider a grammar : $A \rightarrow aB \mid aC$, $B \rightarrow bB \mid b$, $C \rightarrow bC \mid c$.
Design a recursive descent parser. What problem occurs when writing a parser.
3. Construct the deterministic finite automata equivalent to the following NFA.



4. Write short notes on the following :
- Peephole optimization
 - DAG
 - Register allocation and code selection
5. Efficient register allocation and assignment improves the performance of object code-Justify this statement with suitable examples.
6. How a Non-deterministic Finite State Automaton (NFA) differs from a Deterministic Finite State Automaton (DFA).

[2]

7. What is code optimization? Explain about various levels and types of optimizations.

OR

Design a FA (Finite Automata) from given regular expression $10+(0+11)0^*$

8. Write short notes on any two of the following:

- a) Loop optimization
- b) Semantic error
- c) Bootstrapping of compiler
